CHAPTER 2

SEMCD combined with BPS through analog and digital workflow

1 Basic - fabrication steps of a SEMCD BPS denture with analog and digital processes

[1] What should you know before getting started with scanning and designing digital dentures?

For the longest time, the denture fabrication process hasn't seen profound changes until now. But recently, the new invention of the denture fabrication method with CAD/CAM has been efficiently and rapidly developed. Globally, many dental companies have started to create and introduce CAD/CAM manufactured denture methods, and thus it is becoming popular internationally. The Ivoclar digital denture add-on module, together with 3 Shape scanners and software, has already been launched around the world. The most significant advantage of the digital denture manufacturing is that laboratory fabrication material errors, such as the model distortion induced by the stone expansion, the mounting error introduced by the expansion of the mounting stone, denture base resin curing shrinkage and thermal deformation of the master cast can be avoided.

Denture teeth set-up and polished denture surface modifications can be accomplished on the computer within 30-45minutes. Once the starting button is pushed, a PMMA resin disk will be milled automatically. Therefore, the entire denture fabrication processing time is dramatically shortened, and the manufacturing process becomes extremely efficient.

However, even though the laboratory process will change with digital fabrication, the clinical steps, such as impression or bite registration processes, won't change (**Table 2-1**). I insist that while many people regard the digital denture as a panacea for all technical challenges, perfection is still in the clinical details of impressions, bite registration, tooth set-up, resin polymerization and denture delivery. We should make sure that only if these skills are all appropriately performed, we can reap the benefit of digital technology.

I emphasize that inappropriately acquired final impressions or interocclusal records will create ill-fitting and non-functional digital dentures. Therefore, unless the clinician has a high level of clinical skill, this digital technique won't be utilized as effectively as it should be. No matter where the clinician practices dentistry, the highest goal of the clinician should be to "Fabricate the best fitting denture. " Once the concepts of the suction effective dentures and BPS system are incorporated into digital dentistry, more accurate and well-designed CAD/CAM denture will be fabricated in the future.

Digital denture methods only change the laboratory workflow, but the clinical performance remains unchanged

	Analog process	*Digital process*
Preliminary impression	Accu-Dent XD System with Frame cut back tray	Same as the left
Primary bite record	Centric Tray	Same as the left
Mounting	Centric tray bite and horizontal guide onto Stratos articulator	Centric tray bite and UTS CAD onto Virtual articulator
Custom tray	Hand crafted tray with Gnathometer M	Digital manufactured tray with Gnathometer CAD
Mandibular tray design	Based on SEMCD concept	Same as the left
First & second impressions	PVS impression material	Same as the left
Pin-tracing	Gnathometer M	Gnathometer CAD
Tooth setup method	Based on SEMCD/BPS concept, Hand made tooth set-up	Based on SEMCD/BPS concept, Digital tooth set-up
Processing and finishing the dentures	Post-polymerization by IvoBase or Ivocap	Pre-polymerization disks and milled by PM7
Gingiva characterization	SR-Nexco	Same as the left

Table 2-1 Difference between analog and 3 Shape/Ivoclar Vivadent digital denture workflow processes.

2 3shape/Ivoclar Vivadent denture workflow differences between analog and digital process

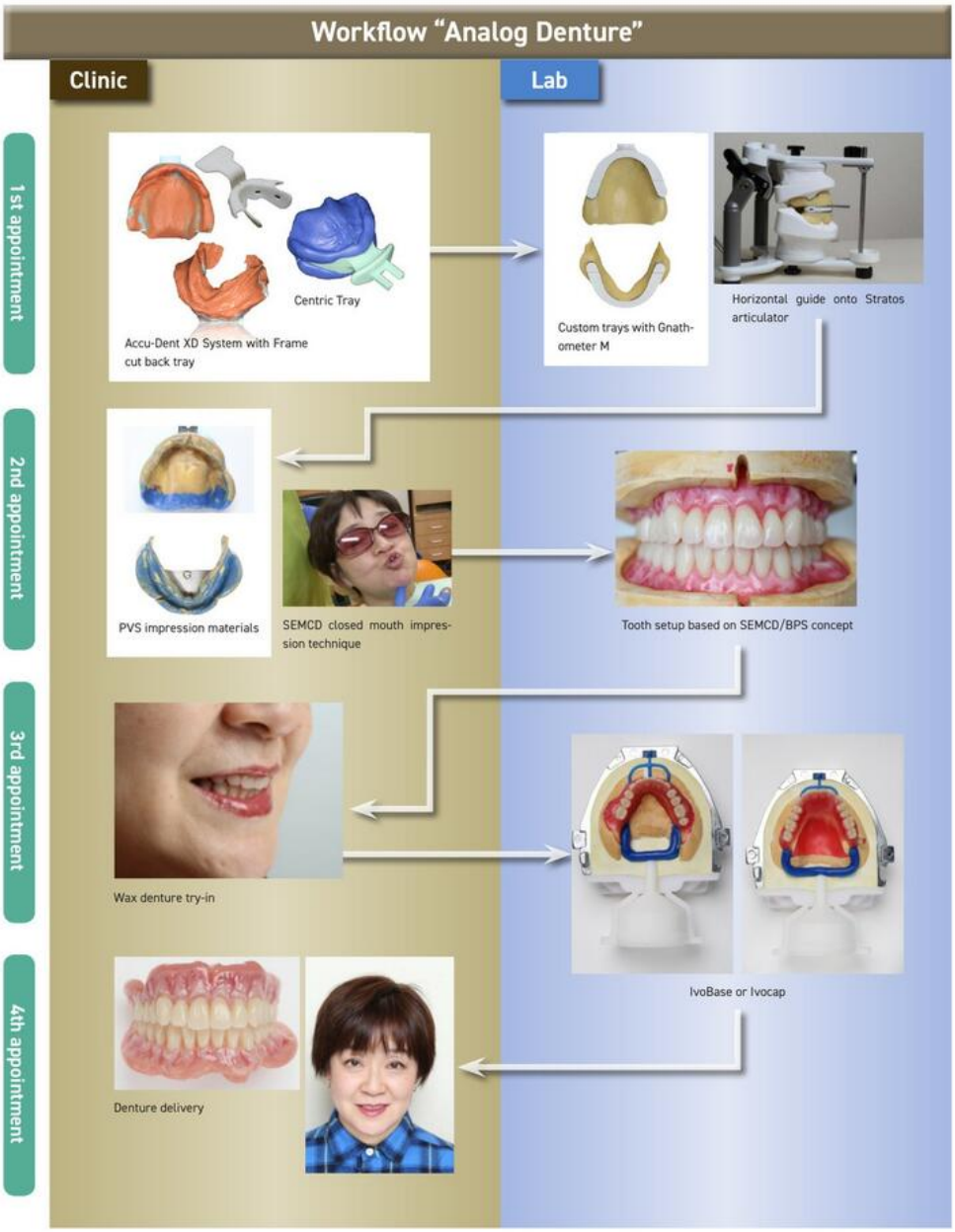


Fig 2-1

Digital denture can change only lab work, but clinical skills are not changed at all

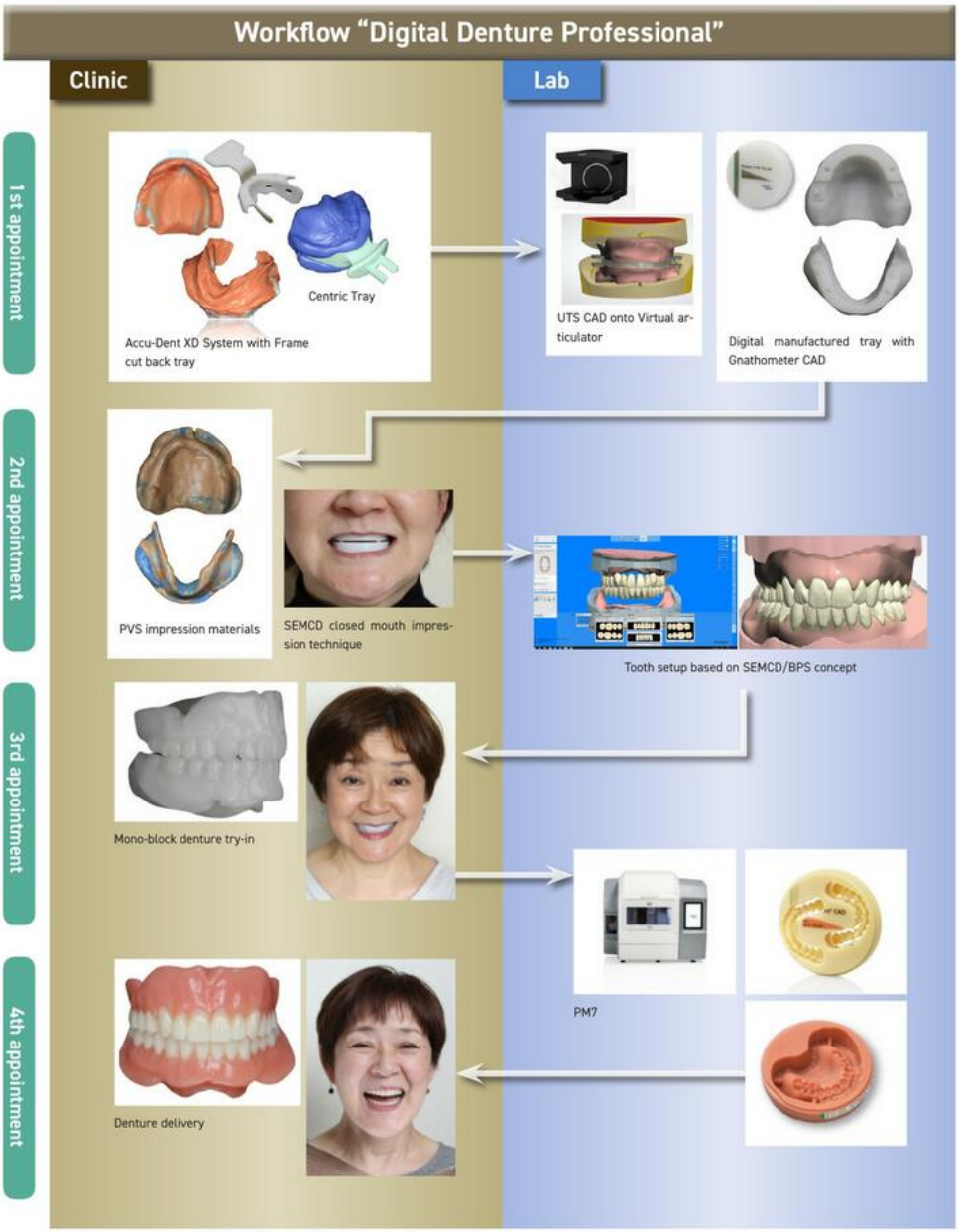


Fig 2-2 (Courtesy for digital pictures by Mr. Mattheus Boxhoorn).